SHIDA WANG (汪铈达)

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GitHub: https://github.com/radarFudan

EDUCATION

B.S. (Math), Fudan University, 2020

Ph.D. (Math), National University of Singapore (NUS), 2024 *Supervisor:* Qianxiao Li

RESEARCH INTEREST

Sequence Modeling, Large Language Model, Recurrent Neural Network, State-space Model

PAPERS

Inverse Approximation Theory for Nonlinear Recurrent Neural Networks (ICLR 2024, spotlight)

StableSSM: Alleviating the Curse of Memory in State-space Models through Stable Reparameterization (ICML 2024)

State-space models with layer-wise nonlinearity are universal approximators with exponential decaying memory (NeurIPS 2023)

LongSSM: On the Length Extension of State-space Models in Language Modelling (ICML 2024 NGSM Workshop)

A Brief Survey on the Approximation Theory for Sequence Modelling (JML 2023)

Efficient Hyperdimensional Computing (ECML 2023)

Integrating Deep Learning and Synthetic Biology: A Co-Design Approach for Enhancing Gene Expression via N-terminal Coding Sequences (ACS Synthetic Biology)

HyperSNN: A new efficient and robust deep learning model for resource constrained control applications (ACC 2026)

DRAFTS

Improve Long-term Memory Learning Through Rescaling the Error Temporally

Learning Relational Tabular Data without Shared Features

Numerical Investigation of Sequence Modeling Theory using Controllable Memory Functions

EXPERIENCE

AI Research Scientist at XTX (2026.06-Now)

AI Residency at XTX (2025.01-2026.05)

Internship at Zhipu (2024.09-12)

Internship at Yi (2024.04-08):

Investigated the length extension for multi-modal setup (needle-in-a-haystack)

Internship at SAIL (2023.04-12):

Investigated the length extension in recurrent language models (state-space models)

Internship at Advance.AI (2021.08-10):

Investigated general anomaly detection techniques such as GAN and Autoencoder.

Teaching Assistant at NUS for DSA5102 (2020.08-11, 2021.08-11)

Internship at Megvii (2019.07-12):

Worked on basic models and Neural Architecture Search.

Internship at Goku Data Limited (2019.01-03):

Worked with daily stock data and tried to produce some new factors

AWARDS AND ACHIEVEMENTS

ICML 2026 silver reviewer, 2026

Neurips 2024 Top Reviewers, Winter 2024

ICML 2024 travel award, Summer 2024

FoS PhD Conference Award (FPCA), Spring 2023

Second Place in Jane Street Electronic Trading Challenge, Summer 2022

First Place in Citadel APAC Regional Datathon, Spring 2021

TALKS

CRUNCH seminar (2024.11.15)

Talk at Fudan University Applied Math PhD Seminar (2024.09.05)

GYSS 2024 Young Scientist Presentation (2024.01.08-2024.01.12)

REVIEW EXPERIENCE

Reviewer for NeurIPS 2024, 2025, ICML 2024, 2025, 2026; AISTAT 2023, 2024, 2025, 2026; HRI 2024, CoLLAs 2024; ACM MM 2024; ECCV 2024, 2026; ICCV 2025; COLM 2024, 2025, 2026; ACM TIST; ACL ARR; ICLR 2025, 2026; TMLR; CVPR 2025, 2026

SKILLS

Fluent in Python (PyTorch, JAX, Triton, TensorFlow), C/C++, Haskell

Familiar with data structure, algorithm, operating system, and parallel programming

RELATED COURSES

Probability, Markov Chain, Brownian motion and Stochastic Calculus, Stochastic Control, Optimal Stopping and Stochastic Control in Finance, Topics in Differential Equations (Fluid Equation), Optimization, Microeconomics, Macroeconomics